NNN-F-289D
February 7, 1977
SUPERSEDING
Fed. Spec. NNN-F-289C
July 18, 1974

FEDERAL SPECIFICATION

FLASK, VOLUMETRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for glass volumetric flasks suitable for laboratory purposes (see 6.1).

1.2 Classification.

1.2.1 Styles, and classes. The volumetric flasks shall be of the following styles, and classes, as specified (see 6.2):

Style 1 - Single capacity line.
Style 2 - Special purpose (see 6.1.2).
Class A - Precision: serialized or unserialized, as specified (see 3.5 and 6.2).
Class B - General purpose.

1.2.2 Capacities. The flasks shall be of the following capacities, as specified (see 6.2):

1 milliliter (ml)	100 ml
2 ml	200 ml
5 ml	250 ml
10 ml	500 ml
25 ml	1,000 ml
50 ml	2,000 ml

1.2.2.1 Availability. Style 1, all classes available in capacities specified in 1.2.2. Style 2, classes available in capacities specified (see 6.1.2 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

FSC 6640

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 148 - Polariscopic Examination of Glass Containers.
E-438 - Glasses in Laboratory Apparatus.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Commercial Standard:

CS 21 - Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

National Bureau of Standards Internal Report:

NBSIR 74-461 - The Calibration of Small Volumetric Laboratory Glassware.

(Application for copies should be addressed to the National Technical Information Service (NTIS), Springfield, VA 22151.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:48Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. Volumetric flasks shall be of borosilicate glass conforming to the requirements of type I, class A of ASTM E 438.

3.2 Annealing. Ring strain shall be no greater than that of temper no. 1 when tested as specified in 4.4.1. Any longitudinal strain, if present, shall be faint and highly diffused.

3.3 Capacity tolerances.

3.3.1 Class A, both serialized and unserialized. Class A volumetric flasks shall contain, as specified, an amount within the capacity tolerance shown in table I.

3.3.2 Class B. Class B volumetric flasks shall contain or deliver, as specified (see 6.2), an amount within the capacity tolerance shown in table I.

3.4 Design and dimensions.

3.4.1 Neck. The neck of the volumetric flask shall be straight, perpendicular to the base, and of circular cross section. The neck shall extend above and below the capacity line or above the highest graduation and below the lowest graduation, an amount shown in table I. Milk glass and/or opaque background surrounding the capacity line(s) or graduation marks, which interfere with the setting, are not acceptable. Style 1 flasks shall have taper-ground joint necks conforming to table I and CS 21. Style 2 necks shall be as specified (see 6.1.2).

3.4.3 Dimensions. Limiting dimensions for style 1 volumetric flasks shall be as shown in table I.

TABLE I. Limiting dimensions for style 1 volumetric flasks

Capacity, ml	± Capacity tolerance, ml		Inside diameter of neck at capacity line, millimeters (mm)		Minimum neck section (mm)				Stopper size
					Above capacity line		Below capacity line		
	Class A	Class B	Minimum	Maximum	Class A	Class B	Class A	Class B	
1	0.01	--	4	5	22	--	5	--	8
2	.015	--	5	6	22	--	5	--	8
5	.02	--	6	8	22	--	5	--	8 or 9
10	.02	0.04	6	8	28	28	7	7	9
25	.03	.06	6	8	35	35	7	7	9
50	.05	.10	6	10	40	40	8	8	9
100	.08	.16	8	12	40	40	10	10	13
200	.10	.20	9	14	50	50	10	10	13 or 16
250	.12	.24	10	15	50	50	10	10	16
500	.20	.40	12	18	60	60	15	15	16 or 19
1,000	.30	.60	14	20	60	60	15	15	19 or 22
2,000	.50	1.00	18	25	60	60	15	15	27

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

3.4.3 Base. The bottom or base shall be such that the volumetric flask will stand solidly on a level surface. The area of the base shall be of sufficient size so that flasks of a capacity larger than 10 ml, when empty and without stopper, will stand on a plane inclined 15° to the horizontal. Flasks with capacities of 10 ml and smaller shall, in a similar manner, stand on a plane inclined 10° to the horizontal.

3.4.4 Stopper. Volumetric flasks shall be furnished with or without a stopper, as specified (see 6.2). If stoppered flasks are furnished, the flask shall be fitted with a standard taper stopper conforming to CS 21 and table I.

3.4.5 Markings.

3.4.5.1 Capacity line. All volumetric flasks shall have a capacity line positioned so as to contain or deliver, as specified (see 6.2), the quantity indicated by the numerical marking. The capacity line shall be a sharply defined line of uniform width extending completely around the circumference of the neck. Within at least two opposite quarters, the line width shall not exceed 0.4 mm or vary more than 0.1 mm. The capacity line shall be parallel to the base and perpendicular to the vertical axis of the neck. The capacity line shall be applied by etching and filling with permanent pigment, by applying a stain which is fired into the glass without etching, or by grit-blasting. There shall be no appreciable change in appearance of marking (see 3.5) when tested as specified in 4.4.3.

3.5 Identification marking. All volumetric flasks shall be permanently and legibly marked with the manufacturer's name or trademark, the capacity, the temperature at which the graduations apply (20°C), and the method of use (to contain or to deliver). Each class A serialized flask shall be marked with a serial number. Unserialized classes A and B flasks may be marked with a serial number, or other identification, at the option of the manufacturer. Class A flasks shall be further marked with the letter "A" to signify compliance with applicable construction and accuracy requirements. Interchangeable ground glass parts (stoppers) shall be marked on both members with the proper standard taper symbol and the size designation in accordance with CS 21. Each volumetric flask shall have an area on one side roughened by blasting or spot-enameled to provide a suitable area for laboratory marking. If spot-enameled, the marking area shall show no loss of enamel (except for dulling) when tested as specified in 4.4.3. There shall be no other change in marking when tested as specified in 4.4.3.

3.6 Workmanship. All flasks shall be transparent and free from striae, surface irregularities, and other imperfections which might distort the appearance of the liquid surface or the portion of the graduation line seen through the glass. The flasks shall be free of any defects which adversely affect their appearance, and shall be free of blisters, boils, seeds, stones, and any other defects which might weaken the glass or impair the serviceability of the flasks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, for which specific inspection has not been provided, have been satisfied.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:48Z
Check the source to verify that this is the current version before use.

4.2 Size of lot. Lot information and size shall be in accordance with MIL-STD-105.

4.3 Sampling and inspection.

4.3.1 Sampling for visual and dimensional examination (end item). Sampling for visual and dimensional examination shall be in accordance with MIL-STD-105, inspection level S-2, and the classification of defects and acceptable quality levels (AQL's) indicated in table II.

TABLE II. Classification of defects for visual and dimensional examination

Categories	Defects	Reference paragraph
Class A, serialized		
Major:	AQL 1.5 percent defective.	
	Neck not straight or perpencidular to base.	3.4.1
	Dimensions not as specified, within tolerances permitted.	3.4.2
	Capacity or graduation lines of wrong width or not straight. (Where lines are filled with stain and the stain extends beyond the edge of the etched line, measure the width of the stain.)	3.4.5.1
	Line or filler breaks degrade a uniform appearance outside the setting arc.	
	Line or filler breaks in setting arc.	
	"A" mark or serial number missing. Marking missing or improper.	3.5
	Chips and cracks (other than obvious shipping damage).	3.6
	Smears of filler, scratches, acid burns, and other blemishes in reading area which interfere with setting.	
	Neck blister: 1 mm on sizes up to and including 250 ml. 1.5 mm on sizes 500 to 2,000 ml.	
	Bulb blister: 2 mm diameter maximum (none if on interior surface).	
Class A, unserialized		
Major:	AQL 4.0 percent defective.	
	Neck not straight.	3.4.1
	Dimensions not as specified, within tolerances permitted.	3.4.2
	Capacity or graduation lines of wrong width or not straight. (Where lines are filled with stain and the stain extends beyond the edge of the etched line, measure the width of the stain.)	3.4.5.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects for visual and dimensional examination (con.)

Categories	Defects	Reference paragraph
Class A, unserialized (cont.)		
Major:	Line or filler breaks degrade a uniform appearance outside the setting arc.	
	"A" mark missing.	3.5
	Marking missing or improper.	
	Chips and cracks (other than obvious shipping damage).	3.6
	Smears of filler, scratches, acid burns, and other blemishes in reading area which interfere with setting.	
	Neck blister: 1 mm on sizes up to and including 250 ml. 1.5 mm on sizes 500 to 2,000 ml.	
	Bulb blister: 2 mm diameter maximum (none if on interior surface).	
Class B, general purpose		
Major:	AQL 4.0 percent defective.	
	Neck not straight or perpendicular to base.	3.4.1
	Dimensions not as specified, within tolerances permitted.	3.4.2
	Capacity or graduation lines of wrong width or not straight. (Where lines are filled with stain and the stain extends beyond the edge of the etched line, measure the width of the stain.)	3.4.5.1
	Marking missing or improper.	3.5
	Chips and cracks (other than obvious shipping damage).	3.6
	Smears of filler, scratches, acid burns, and other blemishes in reading area which interfere with setting.	
	Neck blister: 1 mm on sizes up to and including 250 ml. 1.5 mm on sizes 500 to 2,000 ml.	
	Bulb blister: 2 mm diameter maximum (none if on interior surface).	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

4.3.2 Sampling for tests.

4.3.2.1 Capacity accuracy. Sampling shall be in accordance with MIL-STD-105, inspection level S-4. For class A, an AQL of 1.0 percent defective shall apply. For class B, an AQL of 2.5 percent defective shall apply (see 4.4.2).

4.3.2.2 Annealing and permanency of marking tests. A sample of three flasks shall be taken at random from each 100 flasks, or fraction thereof, in the lot. Each flask in the sample shall be tested for annealing (see 4.4.1) prior to permanency of marking tests. A different 1/3 of the sample shall be used for tests, as specified in 4.4.3.1 and 4.4.3.2. If a flask fails to meet any test listed in this paragraph, the lot shall be rejected.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine whether packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

TABLE III. Examination of preparation for delivery

Examine	Defect
Contents	More or less than specified.
Containers	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers; cushioning inadequate, improper, or missing.

4.3.4 Examination for closure, waterproofing, and banding of containers. When containers are required to be in accordance with PPP-B-566, PPP-B-636, and PPP-B-676, examination for defects in closure, waterproofing, and banding shall be in accordance with the appendix of the appropriate specification.

4.4 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.4.1 Annealing. Flasks shall be tested in accordance with method A or B of ASTM C 148.

4.4.2 Capacity accuracy. Flasks shall be tested for capacity accuracy in accordance with the procedures of NBSIR 74-461.

4.4.3 Stability test. Flasks with a capacity larger than 10 ml shall be placed on a plane inclined 15° to the horizontal. Similarly, flasks with a capacity 10 ml or smaller shall be placed on a plane inclined 10° to the horizontal. The plane surface shall have a sufficient coefficient of friction to prevent the flask sliding. A flask that tips over shall be considered a test failure.

4.4.4 Permanency of marking. A different flask shall be used for each of the following tests:

4.4.4.1 Resistance to alkali. The flask, with stopper removed, shall be completely immersed in 1N sodium hydroxide solution at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the alkali for an additional 1 hour. The flask shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth. For compliance with requirements see paragraph 3.5.

4.4.4.2 Resistance to acid. The flask shall be completely immersed in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the acid for an additional 1 hour. The flask shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth. For compliance with requirements see paragraph 3.5.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:48Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B.

5.1.1.1 Unit container. The number of flasks of one description with or without stoppers (see 3.4.5), as specified (see 6.2) shall be packaged in a box of appropriate size conforming to PPP-B-566, variety I, style 1 or 2, type and class optional; or PPP-B-676, variety I, type, class, and style optional. The box shall be provided with fiberboard locking or cushioning material to prevent shifting of contents. The box closure shall be in accordance with the box specification.

5.1.1.2 Intermediate container. The number of unit containers of flasks of one description, as specified (see 6.2) shall be intermediately packaged in a box conforming to PPP-B-636, class domestic. The box shall be closed to prevent accidental opening.

5.1.2 Level C. Flasks with or without stoppers (see 3.4.5) shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Flasks, packaged as specified in 5.1, in quantities as specified (see 6.2) shall be packed in a box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Flasks, packaged as specified in 5.1, in quantities as specified (see 6.2) shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Flasks, in quantities as specified (see 6.2), packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. Marking of the unit, intermediate, and shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies, or MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use.

6.1.1 Volumetric flasks covered by this specification are intended for laboratory applications.

> Class A serialized volumetric flasks are intended for precision measurements and are of a quality such that specific flasks will be acceptable by the NBS for calibration at specified points if such calibration is justified by the nature of the usage.

6.1.2 Certain specific test procedures require pieces of single-purpose glass apparatus which are essentially modified flasks. Often the quantity of such ware does not justify a detailed specification. In such cases, procurement may be simplified by providing a brief description or sketch referencing this specification and NNN-B-00789, Buret, Straight Precision, for minimum line spacing and length of lines for items which require graduated necks. In apparatus which require enlarged neck sections, the neck shall be straight and cylindrical for at least 1 cm on each side of the marking area. Realistic capacity tolerances can be determined by summing appropriate flask or buret tolerances as applicable to comparative neck diameters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Style, class, and capacity of volumetric flasks required (see 1.2.1, 1.2.2, and 1.2.2.1).
(c) Whether class A precision flasks shall be serialized or unserialized (see 1.2.1 and 3.5).
(d) Whether class B flasks shall contain or deliver (see 3.3.2).
(e) Whether with or without a stopper (see 3.4.4).
(f) Supplementary information as required for style 2 volumetric flasks (see 1.2.2.1, 3.4.1, and 6.1.2).
(g) Selection of appropriate level of packaging and packing (see 5.1 and 5.2).
(h) Marking of document required (see 5.3).
(i) Quantity required (see 5.1.1.1, 5.1.1.2, 5.2.1, and 5.2.2).

Military Activity:

DM

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
HEW-NIH
VA-DMS

PREPARING ACTIVITY: GSA -- FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35¢ each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.